Deep Learning Group Project

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Animal Detection and Counting / 动物检测与计数

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1. Introduction / 引言

This project implements an animal detection and counting system for the Deep Learning final project. The system accepts a single image as input and returns a clean Python-style dictionary whose keys are supported animal categories and whose values are positive integer counts. The final implementation follows the required 20-category vocabulary and writes batch predictions to JSON without any natural-language explanations.

本项目为深度学习期末项目实现了动物检测与计数系统。系统以单张图片作为输入，返回一个干净的 Python 风格字典，键为支持的动物类别，值为正整数计数。最终实现遵循要求的 20 类词汇表，并将批量预测结果写入 JSON 文件，不含任何自然语言解释。

The work is based on YOLO11s and the Ultralytics training/inference framework. YOLO was selected because it performs object localization, classification, and confidence estimation in a single forward pass, which is highly suitable for the time-limited validation and final evaluation workflow. The project covers the complete pipeline: data collection and annotation, model training and fine-tuning, detection inference, post-processing, and JSON prediction export.

本项目基于 YOLO11s 和 Ultralytics 训练/推理框架。选择 YOLO 是因为它在单次前向传播中同时完成目标定位、分类和置信度估计，非常适合有时间限制的验证和最终评估流程。项目覆盖了完整流程：数据收集与标注、模型训练与微调、检测推理、后处理和 JSON 预测导出。

2. Methodology / 方法

2.1 Dataset Construction / 数据集构建

The training data was built progressively through multiple stages. Initially, two source datasets (animal_dataset_xfz and animal_dataset_yzy) were merged and their annotations were converted into a unified JSON format. From this merged dataset, animal_dataset1 was prepared as a 20-class YOLO-format dataset with training and validation splits.

训练数据通过多个阶段逐步构建。首先，合并了两个源数据集（animal_dataset_xfz 和 animal_dataset_yzy），并将其标注转换为统一的 JSON 格式。基于合并后的数据集，准备了 animal_dataset1 作为 20 类 YOLO 格式数据集，包含训练集和验证集划分。

To improve performance on weak categories, two additional targeted datasets were constructed: animal_dataset_5sp_v1 (focusing on wolf, goose, horse, cow, and monkey) and animal_dataset_6sp_v1 (continuing optimization of difficult categories). Both datasets were auto-labeled in LabelMe-style JSON format and then converted to YOLO format while preserving the full 20-class output head. This approach ensured that fine-tuning on targeted categories did not reduce the model's ability to recognize the original 20 animal classes.

为改善弱类别表现，构建了两个额外的定向数据集：animal_dataset_5sp_v1（聚焦狼、鹅、马、牛和猴子）和 animal_dataset_6sp_v1（继续优化困难类别）。两个数据集均以 LabelMe 风格 JSON 格式自动标注，然后转换为 YOLO 格式，同时保留完整的 20 类输出头。该方法确保在定向类别上微调不会降低模型识别原始 20 个动物类别的能力。

- Supported classes: cat, dog, horse, cow, sheep, goat, pig, rabbit, chicken, duck, goose, deer, monkey, fox, wolf, bear, tiger, lion, zebra, giraffe (20 total).
  支持类别：猫、狗、马、牛、羊、山羊、猪、兔、鸡、鸭、鹅、鹿、猴、狐狸、狼、熊、老虎、狮子、斑马、长颈鹿（共 20 类）。
- Data sources include manually provided labels, auto-labeled JSON annotations with visual preview, and targeted fine-tuning datasets for weak classes.
  数据来源包括人工标注、带可视化预览的自动标注 JSON，以及针对弱类别的定向微调数据集。
- Final dataset path: dataset/animal_dataset_6sp_yolo_20cls, configured by configs/animal_dataset_6sp_20cls.yaml.
  最终数据集路径：dataset/animal_dataset_6sp_yolo_20cls，由 configs/animal_dataset_6sp_20cls.yaml 配置。

2.2 Model Training and Fine-tuning / 模型训练与微调

The base detector is YOLO11s (the 'small' variant of YOLO11) with COCO pre-training weights. Training proceeded in three stages, each building on the previous best model:

基础检测器为 YOLO11s（YOLO11 的小型变体），使用 COCO 预训练权重。训练分三个阶段进行，每个阶段基于前一阶段的最佳模型：

Stage / 阶段	Input Data / 输入数据	Purpose / 目的	Output Model / 输出模型
Stage 1	animal_dataset1	Train 20 animal classes from COCO baseline 从 COCO 基线训练 20 个动物类别	animal_dataset1 best.pt
Stage 2	animal_dataset_5sp_v1	Improve wolf/goose/horse/cow/monkey	5sp 20-class best.pt

		改善狼/鹅/马/牛/猴的识别	
Stage 3	animal_dataset_6sp_v1	Continue optimization from best model 从最佳模型继续优化	6sp 20-class best.pt

Training hyperparameters: image size 640×640, batch size 8, freeze first 10 layers, early stopping patience 15 epochs, AdamW optimizer with initial learning rate 0.0005, weight decay 0.0005, 3 warmup epochs. Data augmentations included HSV jitter (h=0.015, s=0.7, v=0.4), rotation (±8°), translation (±10%), scaling (±40%), horizontal flip (50%), vertical flip (5%), mosaic (100%), and light mixup (5%). Mosaic augmentation was disabled after epoch 10. Training was performed on a single NVIDIA GeForce RTX 3070 Ti Laptop GPU using CUDA.

训练超参数：图像尺寸 640×640，批次大小 8，冻结前 10 层，早停耐心值 15 轮，AdamW 优化器（初始学习率 0.0005，权重衰减 0.0005，预热 3 轮）。数据增强包括 HSV 抖动（h=0.015, s=0.7, v=0.4）、旋转（±8°）、平移（±10%）、缩放（±40%）、水平翻转（50%）、垂直翻转（5%）、马赛克（100%）和轻量 mixup（5%）。马赛克增强在第 10 轮后关闭。训练在单张 NVIDIA GeForce RTX 3070 Ti 笔记本 GPU 上使用 CUDA 进行。

2.3 Detection and Post-processing Workflow / 检测与后处理流程

The inference pipeline processes each image through the following steps:

推理流程按以下步骤处理每张图片：

1) YOLO Prediction: Run the fine-tuned YOLO11s model to obtain bounding boxes, confidence scores, and predicted class labels for each detection.

1) YOLO 预测：运行微调后的 YOLO11s 模型，获取每个检测的边界框、置信度分数和预测类别标签。

2) Category Whitelist Filtering: Map YOLO output labels to the 20 supported animal categories. Unsupported labels (e.g., person, bird, elephant) are discarded.

2) 类别白名单过滤：将 YOLO 输出标签映射到 20 个支持的动物类别。不支持的标签（如人、鸟、大象）被丢弃。

3) Confidence Thresholding: Discard detections with confidence scores below 0.25. YOLO's built-in NMS is applied with IoU threshold 0.45.

3) 置信度阈值：丢弃置信度低于 0.25 的检测。YOLO 内置 NMS 以 IoU 阈值 0.45 应用。

4) Custom Duplicate Suppression: An additional post-processing layer beyond YOLO NMS. Same-class boxes are merged when IoU ≥ 0.40 or when centers are close (≤30% of min diagonal) with similar sizes (area ratio ≥ 0.50). Cross-class boxes are merged under stricter conditions: IoU ≥ 0.62 or centers ≤18% of min diagonal with area ratio ≥ 0.70. In all cases, the highest-confidence detection is retained.

4) 自定义重复抑制：YOLO NMS 之外的额外后处理层。同类别框在 IoU≥0.40 时合并，或中心距离近（≤最小对角线的 30%）且尺寸相似（面积比≥0.50）时合并。跨类别框在更严格条件下合并：IoU≥0.62 或中心距离≤最小对角线的 18%且面积比≥0.70。所有情况下保留置信度最高的检测。

5) Counting: Group remaining detections by mapped category name and count instances per category.

5）计数：按映射后的类别名称分组剩余检测，统计每个类别的实例数。

6) JSON Export: Write a dictionary for each image containing only categories with positive counts. Output format: {"eval_000001": {"cat": 2, "duck": 1, "deer": 1}}.

6) JSON 导出：为每张图片写入一个字典，仅包含具有正计数的类别。输出格式：{"eval_000001": {"cat": 2, "duck": 1, "deer": 1}}。

The same duplicate-suppression logic is shared between JSON prediction counting and annotated image visualization, ensuring consistency between displayed boxes and submitted counts.

相同的重复抑制逻辑在 JSON 预测计数和标注图像可视化之间共享，确保显示框与提交计数之间的一致性。

3. Evaluation Result / 评估结果

3.1 Final Model Performance / 最终模型性能

The final model achieved strong performance across all training metrics after the third fine-tuning stage. Epoch 2 was selected as the best checkpoint based on validation mAP50.

最终模型在第三阶段微调后，所有训练指标均表现强劲。基于验证集 mAP50，第 2 轮被选为最佳检查点。

Metric / 指标	Value / 数值
Precision / 精确率	0.963
Recall / 召回率	0.977
mAP50	0.978
mAP50-95	0.940
Best epoch / 最佳轮次	Epoch 2
Model path / 模型路径	runs/detect/.../animal_dataset_6sp_20cls_finetune/weights/best.pt

3.2 Validation Set Results / 验证集结果

The final internal validation used the 15-image validation set (val_set). Using the balanced suppression post-processing configuration, the model achieved an average score of 90.94/100 under the project scoring rules. 9 out of 15 images received perfect scores (100/100), demonstrating that the system correctly handles most detection scenarios.

最终内部验证使用 15 张图片的验证集（val_set）。使用平衡式抑制后处理配置，模型在项目评分规则下取得了

平均 90.94/100 的评分。15 张图片中有 9 张获得满分（100/100），表明系统正确处理了大多数检测场景。

Image Group / 图片组	Representative Results / 代表性结果
Perfect predictions / 完美预测	eval_001384, eval_001388, eval_001431, val_4, val_5, val_6, val_7, val_8, val_10
Main misses / 主要漏检	val_1 missed fox / val_1 漏检狐狸; eval_001391 predicted cow count as 2 instead of 3 / eval_001391 将牛的数量预测为 2 而非 3
Main confusions / 主要混淆	val_2 and val_9 confused duck with goose / val_2 和 val_9 将鸭与鹅混淆; eval_001402 produced extra horse detections / eval_001402 产生多余的马检测

[Image not found: results.png]

[Image not found: eval_001384.png]

4. Analysis / 分析

4.1 Performance Analysis / 性能分析

The model performs excellently on large, visually distinctive animals such as zebra, giraffe, horse, lion, tiger, and bear, where mAP exceeds 0.95. These categories benefit from distinctive shape, texture, and color patterns that YOLO learns well from the training data.

模型在大型、视觉特征明显的动物（如斑马、长颈鹿、马、狮子、老虎和熊）上表现优异，mAP 超过 0.95。这些类别受益于独特的形状、纹理和颜色模式，YOLO 能够从训练数据中很好地学习。

The main remaining errors stem from three sources: (1) visually similar species, particularly duck vs. goose which share body shape and color in synthetic scenes, and fox vs. wolf vs. dog which overlap in texture and pose; (2) occlusion and tightly grouped animals where partial visibility leads to missed or merged detections; and (3) synthetic scene artifacts that may not perfectly represent real-world animal appearances.

剩余错误主要来自三个方面：（1）视觉相似物种，特别是鸭和鹅在合成场景中体型和颜色相似，以及狐狸、狼和

狗在纹理和姿态上重叠；（2）遮挡和密集排列动物，部分可见性导致漏检或合并检测；（3）合成场景伪影可能无法完美代表真实世界的动物外观。

4.2 Error Analysis / 错误分析

Duck/Goose Confusion: These two bird species are the most commonly confused pair. In synthetic images, both appear with similar white/brown body coloration and comparable size, making fine-grained discrimination difficult without more targeted training examples.

鸭/鹅混淆：这两种禽类是最常混淆的组合。在合成图像中，两者呈现相似的白色/棕色体色和相近大小，没有更多定向训练样本的情况下，细粒度区分非常困难。

Canine Confusion: Fox, wolf, and dog share similar body plans, fur textures, and poses. When depicted at a distance or in non-distinctive poses, the model may incorrectly classify among these three categories.

犬科混淆：狐狸、狼和狗具有相似的身体结构、皮毛纹理和姿态。在远距离或非特征性姿态下，模型可能在这三个类别之间错误分类。

Counting Errors: Occlusion and tight grouping remain challenging. When animals partially overlap, the model may detect two animals as one, or one animal as two if body parts are visually separated by occluders.

计数错误：遮挡和密集排列仍然具有挑战性。当动物部分重叠时，模型可能将两只动物检测为一只，或当身体部位被遮挡物视觉分离时将一只动物检测为两只。

False Positive Penalties: The scoring rule deducts 5 points per extra wrong category, making false positive categories especially costly. Conservative confidence thresholding is needed to balance recall against the risk of introducing hallucinated categories.

误报惩罚：评分规则对每个额外错误类别扣除 5 分，使误报类别代价特别高。需要保守的置信度阈值来平衡召回率和引入虚假类别的风险。

4.3 Improvement Strategies / 改进策略

Add more manually verified training examples specifically for duck/goose and fox/wolf/dog pairs, with careful annotation review to ensure labeling consistency.

专门为鸭/鹅和狐狸/狼/狗组合增加更多人工验证的训练样本，并进行仔细的标注审查以确保标注一致性。

Explore test-time augmentation (TTA) during inference to improve robustness, particularly for images with occlusion or unusual viewpoints.

在推理过程中探索测试时增强（TTA）以提高鲁棒性，特别是针对有遮挡或不寻常视角的图像。

Investigate per-class confidence threshold tuning: higher thresholds for frequently confused categories and lower thresholds for reliably detected categories.

研究逐类置信度阈值调优：对经常混淆的类别使用更高阈值，对可靠检测的类别使用更低阈值。

Consider ensemble approaches combining multiple YOLO variants or incorporating additional feature extractors for fine-grained species discrimination.

考虑集成方法，结合多个 YOLO 变体或引入额外的特征提取器进行细粒度物种区分。

5. Contributions / 贡献

The following table describes each team member's contribution to the project.

下表描述了每位团队成员对项目的贡献。

Member / 成员	Contributions / 贡献
Member 1 / 成员 1 待填写 / TBD	Dataset collection, merging, and conversion to LabelMe JSON annotation format. Validation data organization and preprocessing scripts. 数据集收集、合并和 LabelMe JSON 标注格式转换。验证数据组织和预处理脚本。
Member 2 / 成员 2 待填写 / TBD	YOLO model training pipeline, CUDA environment setup, multi-stage fine-tuning, hyperparameter tuning, and model performance tracking. YOLO 模型训练流程、CUDA 环境配置、多阶段微调、超参数调优和模型性能跟踪。
Member 3 / 成员 3 待填写 / TBD	Evaluation and inference scripts, custom duplicate-box post-processing algorithm, annotated image generation, report writing, PPT creation, and error analysis. 评估和推理脚本、自定义重复框后处理算法、标注图像生成、报告撰写、PPT 制作和错误分析。

6. Conclusion / 结论

This project successfully completed a reproducible YOLO-based animal detection and counting pipeline covering the full workflow: data preparation and annotation, multi-stage model fine-tuning, batch inference with custom post-processing, JSON prediction export, validation scoring, and visual annotation with bounding boxes and confidence scores. The final 20-class model achieved a high validation score of 90.94/100 and strong detection metrics (mAP50: 0.978, mAP50-95: 0.940).

本项目成功完成了一个可复现的基于 YOLO 的动物检测与计数流程，覆盖了完整工作流：数据准备与标注、多阶段模型微调、带自定义后处理的批量推理、JSON 预测导出、验证评分以及带边界框和置信度分数的可视化标注。最终 20 类模型取得了 90.94/100 的高验证评分和强劲的检测指标（mAP50: 0.978, mAP50-95: 0.940）。

The key technical contributions include: (1) a progressive data construction strategy that targets weak categories while preserving the full 20-class vocabulary; (2) a balanced duplicate suppression algorithm that reduces false positives without aggressive deletion of true nearby animals; and (3) shared post-processing logic between visual annotation and JSON counting to ensure consistency between displayed and submitted results.

关键技术贡献包括：（1）渐进式数据构建策略，针对弱类别同时保留完整 20 类词汇表；（2）平衡式重复抑制算法，减少误报但不过度删除真实邻近动物；（3）可视化标注和 JSON 计数之间共享后处理逻辑，确保显示结果与提交结果一致。

Future work should focus on collecting more manually verified hard examples for the most confusing category pairs (duck/goose, fox/wolf/dog), exploring per-class threshold optimization, and investigating advanced techniques such as test-time augmentation and ensemble methods to further improve robustness on challenging cases involving occlusion and fine-grained species discrimination.

未来工作应重点关注为最易混淆的类别对（鸭/鹅、狐狸/狼/狗）收集更多人工验证的困难样本，探索逐类阈值优化，并研究测试时增强和集成方法等先进技术，以进一步提高在涉及遮挡和细粒度物种区分的挑战性案例上的鲁棒性。